



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For



No Code and Low Code Agentic AI Applications

TGS Ref No: TGS-2026062147

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 1.1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	17 July 2026	First version — 4-day lesson plan for No Code and Low Code Agentic AI Applications (TGS-2026062147). Topics 1–4 (n8n fundamentals, AI agents, webhook & HTTP request, RAG) adapted from Agentic AI Automation with n8n; Topic 5 human-in-the-loop & guardrails; Topic 6 use cases of agentic AI (banking onboarding, client rapport); Topic 7 voice agents (ElevenLabs & Vapi); Topic 8 mini capstone project and presentation	Tertiary Infotech Academy Pte Ltd
1.1	20 July 2026	Restructured to 3 Learning Units (LU1–LU3) aligned to Course Proposal TSC ICT-BAS-0047-1.1; LOs restated as LO1–LO3 with K/A mapping; added slide-number column to all daily schedules.	Tertiary Infotech Academy Pte Ltd

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Course Title	No Code and Low Code Agentic AI Applications
Course Code	TGS-2026062147
Duration	4 Days (32 training hours)
Daily Schedule	9:30 AM – 6:30 PM (8 training hours/day, excluding lunch)
Lunch Break	1:00 PM – 2:00 PM (1 hour)
Breaks	Short tea breaks are scheduled within each day's training hours
Delivery Mode	Instructor-led, hands-on labs with group work and presentations
Prerequisites	Basic computer literacy; a Google/Gmail account; no coding experience required
Tools & Accounts	n8n (cloud trial + local Docker), OpenAI & Gemini API keys, Telegram, Gmail/Outlook, Google Sheets, Google Calendar, Twelve Data & NewsAPI, ElevenLabs, Vapi

Course Overview

This 4-day hands-on course teaches participants to design, build and secure agentic AI applications on no-code / low-code platforms. Learners set up n8n (cloud trial and local Docker), automate forms and email, store data in Data Tables and spreadsheets, build Telegram-triggered AI agents, integrate webhooks and external APIs, add Retrieval-Augmented Generation (RAG), apply human-in-the-loop and guardrail patterns, implement two real business use cases (retail banking onboarding and a client rapport assistant with human handover), and build voice agents with ElevenLabs and Vapi — culminating in a mini capstone project that is presented and assessed.

Learning Outcomes

By the end of this course, participants will be able to:

Aligned to TSC ICT-BAS-0047-1.1 — Knowledge K1–K6 (assessed by the Written Assessment) and Abilities A1–A5 (assessed by the Practical Performance).

- **LO1 (LU1 · Topics 1–2 · K2, K4 · A1, A2):** Maintain proper configurations of no-code and low-code Agentic AI tools, and describe issues for appropriate escalation.
- **LO2 (LU2 · Topics 3–4 · K3, K6 · A4, A5):** Apply effective prompt engineering, model configuration and Retrieval Augmented Generation to integrate foundational models into low-code AI agent applications.
- **LO3 (LU3 · Topics 5–8 · K5, K1 · A3):** Troubleshoot performance issues in Agentic AI applications using standard debugging methods.

Day 1

Fundamentals of n8n · AI Agents

Morning Session · 9:30 AM – 1:00 PM

Time	Topic / Activity	Duration	Slides
9:30 – 9:45	Welcome, course objectives & overview (TGS-2026062147)	15 min	1–13
9:45 – 12:25	LU1 · Topic 1: Fundamentals of n8n	—	14–103
9:45 – 10:05	Overview of n8n	20 min	16–26
10:05 – 10:35	Set up n8n — create trial accounts (15 learners) and install locally with Docker Compose	30 min	20–22
10:35 – 10:50	Explore the n8n UI	15 min	27–33
10:50 – 11:10	Key concepts of n8n — Triggers, Actions, Nodes, Flows	20 min	34–40
11:10 – 11:35	Activity 1: QR Code form (name, email, phone, message) + email to admin via Gmail/Outlook (2 nodes)	25 min	40–50
11:35 – 12:25	Group Activity (3–4 pax): design an event flyer (e.g. bowling), generate a QR code with the QR generator, build the ad, then present (review past-student examples)	50 min	40–50
12:25 – 1:00	Activity 2: Add an n8n Data Table to Activity 1 to capture submissions (3 nodes)	35 min	40–50
Time	Topic / Activity	Duration	Slides
1:00 – 2:00	Lunch Break	60 min	—

Afternoon Session · 2:00 PM – 6:30 PM

Time	Topic / Activity	Duration	Slides
2:00 – 2:40	Activity 3a: Conditional response — if yes, save the date to the Data Table; if no, send a thank-you email (note: trial Data Table data is not persistent)	40 min	59–66
2:40 – 3:20	Activity 3b: Conditional response — replace the Data Table with Google Sheet/Excel for persistent storage (Google training accounts); if no, send a thank-you email	40 min	67–75
3:20 – 3:35	Tea Break	15 min	—
3:35 – 6:05	LU1 · Topic 2: AI Agents	—	76–103
3:35 – 4:05	Overview of AI Agents — LLM, Memory, Tools and System Instruction	30 min	77–84
4:05 – 4:50	Activity 4a: Telegram-triggered AI Agent — simple customer-service chatbot (OpenAI / Gemini API key provided)	45 min	85–94
4:50 – 6:05	Activity 4b: Telegram-triggered AI Agent (customer service / HR admin) — attach a Data Table as a tool; reuse Activity 2 data or upload a 100-record employee CSV	75 min	95–103
6:05 – 6:30	Day 1 recap, Q&A and wrap-up	25 min	—

Day 2

Webhook and HTTP Request · RAG

Morning Session · 9:30 AM – 1:00 PM

Time	Topic / Activity	Duration	Slides
9:30 – 9:45	Day 1 recap & Day 2 objectives	15 min	—
9:45 – 1:00	LU2 · Topic 3: Webhook and HTTP Request	—	104–150
9:45 – 10:15	Overview of Webhooks — use cases and external triggers	30 min	—
10:15 – 11:35	Activity 5: Website chatbot via webhook — follow the n8n Investment Advisor reference site (CORS, Webhook trigger, Respond to Webhook, AI Agent)	80 min	112–132
11:35 – 11:50	Tea Break	15 min	—
11:50 – 12:10	Selected learners present their live website and chatbot	20 min	—
12:10 – 12:40	Overview of APIs and HTTP Requests — methods,	30 min	—

	headers, query parameters, credentials		
12:40 – 1:00	Activity 6 setup: sign up for Twelve Data and NewsAPI; copy API keys; import the workflow	20 min	133–150
Time	Topic / Activity	Duration	Slides
1:00 – 2:00	Lunch Break	60 min	—

Afternoon Session · 2:00 PM – 6:30 PM

Time	Topic / Activity	Duration	Slides
2:00 – 3:20	Topic 3 (cont.): Webhook and HTTP Request	—	104–150
2:00 – 3:20	Activity 6 (cont.): Configure Twelve Data and NewsAPI keys; build the Finance API → Telegram day-trading agent (candles1min / candles15min / candles1hr + news, AI Agent, Telegram reply)	80 min	133–150
3:20 – 6:30	LU2 · Topic 4: Retrieval-Augmented Generation (RAG)	—	151–186
3:20 – 4:00	Overview of Retrieval-Augmented Generation (RAG) — Tokenization, Embeddings, Vector Stores	40 min	—
4:00 – 4:15	Tea Break	15 min	—
4:15 – 4:45	RAG ingestion walkthrough — the upload → embed → vector-store → retrieve loop; set up OpenAI / Gemini embedding credentials	30 min	—
4:45 – 6:05	Activity 7a: RAG chatbot — upload an IT-Support FAQ PDF from a web page into a vector store, then ask the Telegram bot (Gemini embeddings + knowledge_base tool). Activity 7b: customer-support RAG agent for a training center, ingesting 20 course brochures across three vector databases (Supabase pgvector, Pinecone, Qdrant) answering via a website chat widget	80 min	156–169
6:05 – 6:30	Day 2 recap, Q&A and wrap-up	25 min	—

Day 3

Human in the Loop and Guardrails · Use Cases of Agentic AI

Morning Session · 9:30 AM – 1:00 PM

Time	Topic / Activity	Duration	Slides
9:30 – 9:45	Day 2 recap & Day 3 objectives	15 min	—
9:45 – 1:00	LU3 · Topic 5: Human in the Loop and Guardrails — HR Service Portal	—	187–208
9:45 – 10:25	Overview of Human-in-the-Loop and the HR Service Portal — three coordinated n8n workflows behind one web page (Leave Application, Dashboard, AI Chatbot)	40 min	—
10:25 – 11:25	Activity 8a: Leave Application with Human-in-the-Loop — webhook receives leave request, emails manager an Approve/Reject button (Send and Wait for Response), emails employee the outcome	60 min	189–194
11:25 – 11:40	Tea Break	15 min	—
11:40 – 12:00	Selected learners present their Activity 8a approval flow	20 min	189–194
12:00 – 12:25	Overview of Guardrails — pre/post patterns, blocked responses, canned-reply fallbacks	25 min	—
12:25 – 1:00	Activity 8b & 8c: Build the HR Dashboard Data webhook (leave-balance stats) and the AI Chatbot with Input & Output Guardrails; wire all three endpoints in the HR Service Portal	35 min	195–201
Time	Topic / Activity	Duration	Slides
1:00 – 2:00	Lunch Break	60 min	—

Afternoon Session · 2:00 PM – 6:30 PM

Time	Topic / Activity	Duration	Slides
2:00 – 6:30	LU3 · Topic 6: Use Cases of Agentic AI	—	209–233
2:00 – 2:20	Overview — agentic AI in the workplace; the 4-step problem-solving framework; agent decisions vs deterministic steps	20 min	—
2:20 – 2:55	Activity 9 Part A (group): retail banking onboarding — problem definition, root cause analysis, solution ideation (Marina Trust Bank)	35 min	213–222
2:55 – 4:15	Activity 9 Part B: build the onboarding AI Agent (duplicate check, KYC screening, eligibility rules, customer record, email) — form version, then the bank-website webhook version	80 min	213–222
4:15 – 4:30	Tea Break	15 min	—
4:30 – 5:45	Activity 10: client rapport assistant with human handover (Meridian Asset Management) — agent drafts a non-advisory reply; a relationship manager approves or declines before anything is sent; three-tab audit trail	75 min	223–233
5:45 – 6:05	Selected learners present their use-case builds	20 min	—
6:05 – 6:30	Day 3 recap, Q&A and wrap-up	25 min	—

Day 4

Voice Agents · Mini Capstone Project and Presentation

Morning Session · 9:30 AM – 1:00 PM

Time	Topic / Activity	Duration	Slides
9:30 – 9:45	Day 3 recap & Day 4 objectives	15 min	—
9:45 – 1:00	LU3 · Topic 7: Voice Agents	—	234–259
9:45 – 10:15	Overview of Voice Agents — ElevenLabs vs Vapi architectures; signed URLs; public vs private keys; tool webhooks	30 min	—
10:15 – 12:15	Activity 11: Voice booking agent with ElevenLabs (GG Hair Salon) — signed-URL flow, Google Calendar tool webhooks, knowledge-base grounding, end-to-end voice booking	120 min	238–248
12:15 – 12:30	Tea Break	15 min	—
12:30 – 1:00	Activity 12 setup: import the Vapi Custom-LLM flow, prove the webhook with curl, read the guardrail prompt	30 min	249–258
Time	Topic / Activity	Duration	Slides
1:00 – 2:00	Lunch Break	60 min	—

Afternoon Session · 2:00 PM – 6:30 PM

Time	Topic / Activity	Duration	Slides
2:00 – 3:00	Topic 7 (cont.): Voice Agents	—	234–259
2:00 – 3:00	Activity 12 (cont.): grounded FAQ voice agent with Vapi (MediRefill) — create the assistant, wire the public key, run the graded calls (grounded answer, medical refusal, emergency escalation)	60 min	249–258
3:00 – 6:30	LU3 · Topic 8: Mini Capstone Project and Presentation	—	260–268
3:00 – 3:20	Capstone briefing — requirements, scope and assessment criteria	20 min	263–268
3:20 – 4:35	Mini capstone build (supervised lab time)	75 min	260–268
4:35 – 4:50	Tea Break	15 min	—

4:50 – 5:45	Presentation of mini capstone projects	55 min	260–268
5:45 – 6:30	Assessment (WA SAQ + PP briefing) and course closing	45 min	263–268

Tools, Accounts & Resources

QR Code Generator	https://alfredang.github.io/qrcodegenerator/
Webhook reference — Activity 5: Investment Advisor	https://alfredang.github.io/n8n-investmentadvisor/
Finance API reference — Activity 6: Finance Advisor	https://alfredang.github.io/n8n-financeadvisor/
Finance Data API — Twelve Data	https://twelvedata.com/login
News API — NewsAPI	https://newsapi.org/
LLM API keys	OpenAI API key & Google Gemini API key (provided to learners)
Voice — ElevenLabs	https://elevenlabs.io/ (Conversational AI agent + API key)
Voice — Vapi	https://vapi.ai/ (assistant + public key)
Local n8n	Docker Compose self-hosted install (see Learner Guide)

Assessment

Participants are assessed through a Written Assessment (SAQ, 1 hour) and a Practical Performance assessment (1 hour), plus hands-on activities across the four days and a mini capstone project presented on Day 4. Work is evaluated against the course learning outcomes — workflow design, AI agent integration, webhook and API use, Retrieval-Augmented Generation, human-in-the-loop and guardrails, business use cases, and voice agents.